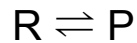


Consider the reaction:



Reaction kinetics data with and without enzyme E are shown in the following table:

|                  | Substance | Concentration (mM) |       |        |         |           |
|------------------|-----------|--------------------|-------|--------|---------|-----------|
|                  |           | Initial<br>(0 min) | 1 min | 10 min | 100 min | 1,000 min |
| No enzyme        | R         | 300                | 300   | 300    | 300     | 300       |
|                  | P         | 20                 | 20    | 20     | 20      | 20        |
| With 5 nM enzyme | R         | 300                | 263   | 100    | 50      | 50        |
|                  | P         | 20                 | 57    | 220    | 250     | 250       |

Based on the data, which of the following statements are accurate?

- I. The enzyme-catalyzed reaction has reached equilibrium by 100 minutes.
  - II. The reaction with no enzyme begins the experiment at equilibrium.
  - III. The enzyme–transition state complex has lower free energy than either R or P.
  - IV. The  $\Delta G$  between the transition state and the product P is smaller in the presence of enzyme than without.
- ✔ ☒ A. I and IV only [49%]
- ☐ B. II and III only [7%]
- ☐ C. I, II, and III only [27%]
- ☐ D. I, II, III, and IV [14%]

Correct

49% answered correctly

## Explanation:

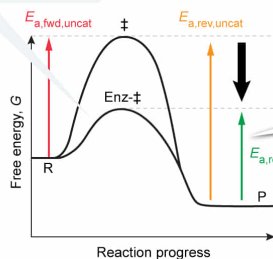
### Enzymes lower activation energy but do not affect equilibrium

(Number II) Without an enzyme, concentrations are constant due to large  $E_{a, fwd, uncat}$ , not due to equilibrium.

|                  | Substance | Concentration (nM) |       |        |         |           |
|------------------|-----------|--------------------|-------|--------|---------|-----------|
|                  |           | Initial (0 min)    | 1 min | 10 min | 100 min | 1,000 min |
| No enzyme        | R         | 300                | 300   | 300    | 300     | 300       |
|                  | P         | 20                 | 20    | 20     | 20      | 20        |
| With 5 nM enzyme | R         | 300                | 263   | 100    | 50      | 50        |
|                  | P         | 20                 | 57    | 220    | 250     | 250       |

(Number III) The enzyme lowers the  $G$  of the transition state ( $\ddagger$ ), but it remains higher than free R and P.

(Number I) Reaction proceeds then stops, due to equilibrium.



(Number IV) Enzymes lower  $E_a$  ( $\Delta G$  to each transition state) for both the forward and reverse reactions (ie, from product).

Enzymes are biological catalysts that **speed up** chemical reactions by **binding and stabilizing** the reaction's **transition state ( $\ddagger$ )**, forming a short-lived enzyme–transition state **complex**. By stabilizing the transition state, enzymes lower the **activation energy** of the forward reaction, which is the free energy difference between the transition state and reactants (ie,  $\Delta G_{\ddagger-\text{reactants}}$ ). The **transition state is the same** for the forward and reverse reaction; therefore, the free energy difference between the transition state and the products (ie,  $\Delta G_{\ddagger-\text{product}}$ ) is also **lowered** in the presence of enzymes (**Number IV**). Without enzymes, many biological reactions are **very slow** and may not appear to proceed at all.

Through catalysis, enzymes allow reactions to reach **equilibrium** more quickly; however, equilibrium itself is **not affected** by the enzyme's presence. Equilibrium occurs when the forward and reverse reactions rates **equal** each other. For catalyzed reactions, this can be recognized when product and reactant concentrations **stop changing**, allowing the

[products] to [reactants] ratio (ie, the **reaction quotient  $Q$** ) to reach a constant value (ie, the **equilibrium constant  $K_{eq}$** ).

In this question, the enzyme-catalyzed reaction **ceases changing concentrations** after 100 minutes have passed, indicating **equilibrium has been reached** with  $K_{eq} = 250 \text{ mM} / 50 \text{ mM} = 5$  (**Number I**).

**(Number II)** The enzyme-catalyzed data establish the value of the equilibrium constant,  $K_{eq} = 5$ , and this value *does not change* whether the enzyme is present or not. However, the uncatalyzed reaction has a [product] to [reactant] ratio of 20 mM : 300 mM (ie,  $Q = 0.067$ ), and so is *not* at equilibrium. The R and P concentrations do not appear to change over the course of the experiment due to the high activation energy, which causes the reaction to be immeasurably slow.

**(Number III)** Although the free energy of the enzyme–transition state complex is lowered, it is still higher than the energy of either free reactant or product. If the complex had the lowest free energy, the substrate would remain bound and would not be released.

### **Educational objective:**

Enzymes are biological catalysts that speed up biochemical reactions by lowering the free energy of the transition state, and thus lowering the energy of activation. Enzymes do not alter the free energy of either free reactant or product.

**Time spent:**0 sec

**Subject:**  
Biochemistry

**QID:**403705

**System:**  
Enzymes